

To harvest or not to harvest? Forest management as a trade-off between bioenergy production and carbon sink

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ABSTRACT

Due to global warming and energy conservation, the importance of forests should be seriously considered. Forest could not only produce bioenergy by harvest to substitute fossil fuels, but also sequester carbon dioxide from the air without harvest. How to measure the trade-off between forests as bioenergy production and carbon sinks is the aim of this paper. The forest management for the trade-offs is modelled theoretically based on two perspectives of carbon reductions and benefits. The magnitude of trade-offs is calculated considering carbon reduction, ecological benefit, and economic benefit by empirical analysis based on the forestry database of 180 countries. The empirical results reveal that the magnitude of trade-offs is lowest in Iran both between carbon sinks and carbon flow from the perspective of carbon reductions and that between economic benefit and ecological benefit based on forest income. It implied the coordinated development of wood fuel and carbon sinks in Iran's forest management. Meanwhile, the magnitude of trade-offs is highest in Ecuador between carbon sinks and carbon flow, and it is highest in Togo between economic benefit and ecological benefit. It showed that forest management should be improved. Furthermore, energy and environment policy implications could be proposed based on forest management. Effective forest management should be considered to help fully realize energy conservation and environment protection.

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1. Introduction

In response to global climate change and energy conservation, countries around the world are seeking effective strategies to balance the relationship between the economy and the environment. Forestry is an important economic and environmental asset. Forests can produce carbon sinks without the harvests and bioenergy generated by wood products, substituting fossil fuels with the harvests. Therefore, two important ways for forests to mitigate climate change are bioenergy generated by wood fuel of forest and carbon sinks. The trade-off between the two for forest management is pivotal. Some activities are been carried out, such as afforestation and reforestation activities under the Clean Development Mechanism. This mechanism allows developed countries to transfer and

acquire project-level emission reduction offsets, thereby implementing greenhouse gas reduction projects in developing countries. The project must pass strict monitoring and certification to take effect. It can effectively help forest carbon sinks achieve market value to realize the economic and environmental benefits. Developing countries with richer forest resources could obtain an amount of economic income while developed countries should pay for it. Therefore, combining the fixed carbon dioxide capacity of forests with carbon trading, and developing into a forestry carbon sink, do not only deal with climate change, but also bring economic benefits to local forest resource-rich areas. According to the 2017 *Global Bioenergy Statistics Report* launched by the World Biomass Energy Association (WBA), forestry, remains a key part of biomass supply, accounting for 87% of total biomass supply, including firewood, wood waste, recycled wood, charcoal and so on. Therefore, forestry is a major source of biomass energy supply and plays an important role in the development of renewable energy.

For the harvest, the forest could help reduce carbon emissions for producing bioenergy, which is a cleaner alternative to fossil energy, there are also some other forest products with economic

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benefits. The study of Jästad et al. (2019) showed that the biofuel implementations based on forests had the economic spillover effects on other parts of the forest sector. The use of bioenergy can reduce greenhouse gas emissions and increase energy security (Hagberg et al., 2016). Therefore, the sustainable bioenergy supply chain has been redesigned to analyze how to achieve economic, environmental and social goals (Rabbani et al., 2018; Kraxner et al., 2016). Bioenergy should be considered in making regional and global greenhouse gas mitigation policies (Wise et al., 2015). Some studies had investigated the status and management of bioenergy in different countries, such as in the US (Smith et al., 2018), England (Townsend et al., 2018), Norway (Cavicchi, 2018), Egypt (Kamel et al., 2018) and so on. Deboni et al. (2019) proposed models to calculate the price of forest biomass used as bioenergy. Furthermore, Keller et al. (2018) attempted to estimate the cost and emission benefits to the electrical system with the conversion of coal to bioenergy that is made from forest residue in Alberta. And Alsaleh and Abdul-Rahim (2018) analyzed the macroeconomic determinants of the cost efficiency rate in the bioenergy production among developing and developed members of the EU28 countries. From the global perspective, Jin et al. (2018) evaluated the combined global implications of regional bioenergy expansion effort which was supplied from forest biomass. Johnston and van Kooten (2016) examined the effects of expanded demand for wood to generate electricity by using a static global forest trade model of wood products.

For forest standing biomass without a harvest, the forest is a huge carbon sink and it is an alternative way to mitigate climate change (Li et al., 2018; Klapwijk et al., 2018). The strategies for mitigating climate change could be summarized as carbon emissions reduction and carbon sequestration. From the perspective of carbon emissions reduction, energy-saving potential was estimated (Lin and Moubarak, 2014) and inter-fuel substitution possibilities were evaluated (Lin and Wesseh, 2013), and other researches had been done. For carbon sequestration, the cost efficiency and economic effects of forests used for carbon emissions reduction were analyzed (Yuen et al., 2017; Sloan et al., 2018; Hong et al., 2018). Hadden and Grelle (2016) show that managed forest area is a carbon sink. Globally, forests may increase carbon sequestration through biologically-mediated silicate weathering by about 32% (Song et al., 2018). Peltoniemi et al. (2006) analyzed the decisive factors, by calculation of the forest carbon budget in Finland. Moreover, Shu et al. (2019) found that environmental factors have an impact on the forest carbon sink capacity. However, there are also some obstacles that policies of mitigating climate change are linked to managed forests (Klapwijk et al., 2018). Lubowski et al. (2006) examined the evidence for implications of forest which include carbon benefits, environmental and monetary costs, and the tradeoff between different strategies in the United States.

Forests have a close relationship with the climate in terms of carbon sink and emissions (Geng et al., 2017). Forest biomass has significant potential to mitigate climate change by using it for energy and sequestration of carbon in new biomass (Bentsen, 2017). Ni et al. (2016) did a research around the world which shows that carbon flow of forests harvest is beneficial for carbon reductions and carbon stocks in standing forest biomass have a potential for storage of carbon dioxide. However, the two strategies cannot be achieved at the same time. Hence forest management strategies should be integrated with energy, forest products and land management policies (Malmsheimer et al., 2011). The trade-offs between the objects which includes timber production, carbon stocking and habitat quality should be measured the possible outcomes (Carpentier et al., 2017). However, it is possible to reduce the trade-offs between the objectives of forest management (Triviño et al., 2017). In this way, decision-makers of forest

management should evaluate how to choose bioenergy usage as the most beneficial for carbon emission reduction goals for the short-term, medium or long term (Zanchi et al., 2012). Eriksson et al. (2018) found when the social planner does not recognize the emissions and sequestration from the forest, the planner will set bioenergy levels that are too high and the levels of forest standing biomass too low. However, Ni et al. (2016) suggested that increasing harvest over the next 100 years would increase carbon sinks and there would be no conflicting between carbon sequestration and wood production. Tatoutchoup (2016) used a dynamic land allocation model with the infinite rotation to determine the recycling rate, which could maximize the forest area, thereby it can also maximize the social net benefit. Forest management strategies have different effects on socioeconomics and other ecosystem services (Spies et al., 2017). And there are unmanaged forests not being valued by carbon fixation measurements which also should be considered (Krug et al., 2012). Forests have an impact on carbon storage and the effects vary with the management strategies (Kalies et al., 2016).

Therefore, the magnitude of trade-offs between bioenergy production and carbon sink is worthy of study around the world. The value of forestry in this paper is mainly summarized into the following three items: First value is the forest carbon sinks which could fix carbon dioxide from the air. The second value is the production of biomass fuel such as fuelwood, which could substitute the use of fossil fuels, thereby reducing carbon dioxide in the air. The last one is forest products can bring economic benefits. This study has two innovations: one is to analyze the trade-offs between using biomass to displace fossil resources or using biomass to store carbon dioxide. And theoretical models with introducing carbon price are established from ecological and economic perspectives which are total carbon reductions and total incomes. The other one is to calculate the magnitude of trade-offs between bioenergy production and carbon sinks based on the database of 180 countries. In addition to tree planting, there is no other cost for forest carbon sequestration. However, forests as a source of bioenergy have some other costs. Hence this paper does empirical analysis from two perspectives of the total carbon reduction and forestry income (ecological benefits and economic benefits).

2. Theoretical models

2.1. Carbon stock and flow

Forests play an important role in sequestering carbon dioxide and the main sources of biomass energy. Bioenergy combustion can replace fossil fuels whose combustion will produce plenty of carbon dioxide. The replacement of fossil energy by bioenergy will reduce carbon dioxide emissions. According to the International Energy Agency, biomass energy will account for 50% of global energy consumption by 2050. Harvesting can make carbon stocks in forests flow into fuel wood which is one of the forests' final products. Therefore, forest is a source of bioenergy and considered to be a carbon sink.

Forest management is expected to maximize carbon sequestration and carbon reduction of replacing fossil fuel by bioenergy. The trade-off between the two strategies should be considered. To analyze the total carbon reduction based on forest and the tradeoff between the carbon sinks and bioenergy, a model is constructed by following Pingoud et al. (2018). There are two assumptions considered here. The first assumption is a unit forest is composed of a large number of stands with a uniform age-class distribution for which a constant management regime to make the results more comparable and the second is that clear-cutting cycle is applied repeatedly.

In specific forest areas, forests are cut down at time t , and felled trees can be used for the production of forest products, including bioenergy and other forest products. The increase in biomass after harvesting compared to time t_0 is the amount of forest carbon storage that can fix carbon dioxide in the air. Therefore, there are two forms of the increase in forest carbon at time t_1 relative to time t_0 . The one is carbon flows that forests being converted to bioenergy substituting fossil fuel and the other one is carbon sinks which are increments of forest carbon stocks. At the time t_2 , forest management repeats the production of bioenergy and carbon sinks in this forest area as the same as t_1 (As shown in Fig. 1). Different forest areas have different management systems, so the production of bioenergy and the amount of carbon sinks formed each year are different. The total carbon reductions at time t include the conversion to bioenergy to substitute fossil fuels and carbon storage with increased forest carbon stocks.

According to the analysis, the equations of the relationship between carbon stock and flux can be express as follow.

$$C_t = C_SINK_t + C_BIO_t \quad (1)$$

$$C_BIO_t = H_BIO_t * DF_{bio_fossil} \quad (2)$$

$$H_t + IM_t - EX_t = H_BIO_t + H_MAN_t \quad (3)$$

C_SINK_t and C_BIO_t represent the carbon emissions fixed by forest carbon sinks and carbon emissions reduction by fossil fuel substitution with bioenergy. C_t is the volume of forest biomass which can make carbon sequestrated from the atmosphere. H_t , H_BIO_t and H_MAN_t are the harvest, harvest for bioenergy and harvest for other forest products. IM_t and EX_t are the number of imports and exports. DF_{bio_fossil} is the displacement factor that the harvested carbon allows for an annual proportional reduction in fossil carbon emissions into the air.

2.2. Economic benefit and ecological benefit

Forestry has two types of benefits, benefit in currency is from the forestry manufactured products and bioenergy produced by biological material, the other benefit is from environmental improvement. The environmental benefit can be valued as the

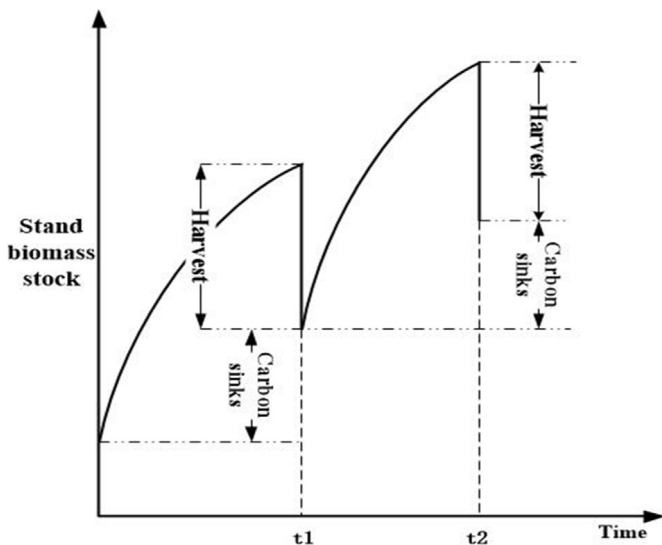


Fig. 1. Forest harvest and carbon sinks over time in a management regime.

carbon reduction in the atmosphere through the forest carbon sinks and reduction in fossil fuel consumption. This article assumes that forest in a specific country can be represented by an average forest stand which is a simplification compared to other forest growth models, but it can be a reasonable assumption in our case since the analysis is from the country aspect globally (Vass and Elofsson, 2016; Hoel and Sletten, 2016).

The amount of forest standing biomass volume is represented by BV_t^i in country i , at time t and the forest biomass volume in period $t-1$ is BV_{t-1}^i where $i=1, \dots, I$ and $t=0, 1, \dots, T$. BV_t^i is determined by the amount of forest biomass volume, the change in the volume of the forest, BV_t^i , and the harvest, H_t^i at time t . All the variables are accounted for at the end of the year. Above contents can be described in the following equation, the forest biomass BV_0^i is assumed as a constant at the time 0.

$$BV_t^i = BV_{t-1}^i + BV_t^i - H_t^i \quad (4)$$

The harvest is used for the production of bioenergy, BE_t^i , and the forestry manufacture, M_{jt}^i . Besides, there are quantities of forest products being exported and imported. It can be represented by EX_t^i and IM_t^i , respectively. j represents forestry manufactured products. Forest products include sawn wood, plywood, veneer sheets, particleboard, fiberboard, mechanic pulp, chemical pulp, newsprint, printing and writing paper and so on (FAO, 2015).

$$H_t^i + IM_t^i - EX_t^i = BE_t^i + \sum_{j=1}^J M_{jt}^i \quad (5)$$

Therefore, forest carbon sequestration is the incremental carbon content in forest biomass in the period t , it can be expressed by equation (6).

$$CS_t^i = BV_t^i - H_t^i \quad (6)$$

The quantity of carbon sequestered is determined by incremental forest biomass. According to the earlier researches, the average sequestration coefficient is denoted by α . When bioenergy substitutes fossil fuels, this carbon emissions offset is denoted as γ and expresses the number of tons of fossil fuel carbon replaced by one ton of carbon in bioenergy. There is consideration of the technologies that are used in the harvesting and conversion of the biomass. The emission from the technologies used for harvesting, transporting and processing forest products and bioenergy is assumed as the percentage of the biomass and is denoted as ϕ . Hence the total carbon reduction, CR_t^i , is calculated by equation (7).

$$\begin{aligned} CR_t^i &= \alpha CS_t^i + \gamma BE_t^i - \phi H_t^i \\ &= \alpha (BV_t^i - H_t^i) + \gamma BE_t^i - \phi H_t^i \end{aligned} \quad (7)$$

The total forestry benefit can be calculated by the sum of the earnings from the products market and the carbon market.

$$\begin{aligned} TB_t^i &= TB_EM_t^i + TB_EL_t^i = \left[P_{BE_t^i} - C_{BE} (BV_t^i) \right] * BE_t^i + \sum_{j=1}^J M_{jt}^i * \\ &\quad \left(P_{M_{jt}^i} - C_{M_{jt}^i} \right) + P_{CARBON_t} * \left[\alpha (BV_t^i - H_t^i) + \gamma BE_t^i - \phi H_t^i \right] \end{aligned} \quad (8)$$

$TB_EM_t^i$ and $TB_EL_t^i$ represent the forest economic benefit and ecological benefit, respectively. $P_{BE_t^i}$, P_{CARBON_t} are the prices of the bioenergy and carbon per unit at time t . The bioenergy cost,

$C_{BE}(BV_t^i)$, is assumed to vary as a function of the stocking biomass, economic wealth and population density (Obersteiner et al., 2006). P_{Mjt}^i and C_{Mjt}^i denote the price and cost of the j th forestry manufactured product in the country i , at time t .

During the period from the base year to the T year, the total benefit is represented by the following equation, $\rho = \frac{1}{(1+r)}$, r is the discount rate.

$$TB^i = \sum_{t=0}^T \rho^t * TB_t^i$$

$$= \sum_{t=0}^T \rho^t * \left\{ \left[P_{BE_t}^i - C_{BE}(BV_t^i) \right] * BE_t^i + \sum_{j=1}^J M_{jt}^i * \left(P_{Mjt}^i - C_{Mjt}^i \right) + P_{CARBON_t} * \left[\alpha (\dot{BV}_t^i - H_t^i) + \gamma BE_t^i - \phi H_t^i \right] \right\} \quad (9)$$

The forest management problem becomes to maximize the total benefit,

$$\max_{BE_t^i, BV_t^i, H_t^i} TB^i \quad (10)$$

Subject to equations (4)–(7).

The discrete-time Lagrangian could be established to solve the problem. From the first-order conditions, the steady-state solutions are:

$$\lambda = P_{BE}^i - C_{BE}^i + P_{CARBON} * \gamma \quad (11)$$

$$\lambda = P_M - C_M \quad (12)$$

$$\mu = P_{CARBON} * (\alpha + \phi - \gamma) - (P_{BE}^i - C_{BE}^i) \quad (13)$$

$$\dot{BV}^i = H^i \quad (14)$$

$$CS^i = 0 \quad (15)$$

$$BV^i = BV_0^i \quad (16)$$

$$BE^i = \left[P_{CARBON} * (\alpha + \phi - \gamma) - P_{BE}^i - C_{BE}^i \right] * (1 - \rho) / C_{BE}^i \quad (17)$$

From the steady-state solutions, it shows Lagrange factor λ is the shadow price of the bioenergy and the manufactured products. And the benefit of the bioenergy is equal to that of the forest products. It means that forest products and bioenergy are perfect substitutes in production. This result is derived from a simplification model which simplifies the proceedings of bioenergy and manufactured products since forest products can more easily be used as bioenergy than bioenergy substrate can be used as forest products. All forest products can be incinerated to be used as bioenergy, but some of the bioenergy substrate, such as harvesting residues, cannot be used for forest products. The result is similar to the share of harvest used for bioenergy which varies considerably between countries (Vass and Elofsson, 2016). And the result of forest carbon sinks is consistent with Hedenus and Azar (2009), their study illustrated that under strict climate policies, long-term rotation for carbon sequestration will not be cost-effective in the long run. While Eriksson (2015) found that increasing forest biomass is more conducive to reducing carbon in the atmosphere than increasing forest bioenergy globally. The reason for the deviation is that this

study gives the advantage of using bioenergy instead of fossil fuels to reduce carbon emissions. μ represents the environment cost of the forest harvested for the production of bioenergy. The incremental forest biomass volume is all harvested and used for bioenergy and forest products in a steady state. Hence, the biomass volume is still at the level of the base year. However, the results from the theoretical analysis may not be consistent with the harvest situations in all the countries worldwide. Furthermore, this paper uses data from 180 countries to analyze different forest management on how to trade-off.

2.3. Calculation of trade-offs

The magnitude of trade-offs between the bioenergy production and carbon sinks can be measured by the root mean square deviation (RMSD). Root mean square deviation commonly used to measure the magnitude of trade-offs between the paired indicators of ecosystem service (Bradford and Amato, 2012; Feng et al., 2017; Zeng et al., 2019). This paper is to measure the trade-offs between bioenergy production and carbon sinks, thus the indicators are the carbon stock, carbon flow, forest economic benefit, and ecological benefit. Based on the analysis of the above sections, RMSD can be used to measure the magnitude of the trade-offs of two paired indicators which are carbon sinks (FM_1) versus carbon flow (FM_2) and forest economic benefit (FM_3) versus ecological benefit (FM_4).

$$FM_1 = C_SINK_t \quad (18)$$

$$FM_2 = C_BIO_t = H_BIO_t * DF_{bio_fossil} \quad (19)$$

$$FM_3 = TB_EM_t^i = \left[P_{BE_t}^i - C_{BE}(BV_t^i) \right] * BE_t^i + \sum_{j=1}^J M_{jt}^i * \left(P_{Mjt}^i - C_{Mjt}^i \right) \quad (20)$$

$$FM_4 = TB_EL_t^i = P_{CARBON_t} * \left[\alpha (\dot{BV}_t^i - H_t^i) + \gamma BE_t^i - \phi H_t^i \right] \quad (21)$$

The indicators should be first standardized to make the unit differences eliminated among the different indicators.

$$FM_{std} = \frac{FM_{obs} - FM_{min}}{FM_{max} - FM_{min}} \quad (22)$$

FM_{std} is the standardized value of the given indicator where it is carbon stock, carbon flow, economic benefit and ecological benefits. And the FM_{obs} is the observed value, FM_{min} is the minimum observed value, FM_{max} is the maximum observed value. Then the root mean square deviation (RMSD) can be calculated as follow:

$$RMSD = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (FM_i - FM_{exp})^2} \quad (23)$$

FM_i is the standardized value of the given indicator and FM_{exp} is the expected value of the paired indicators. Therefore, the RMSD represents the departure from the 1:1 line of the equal benefit. And the larger the value of RMSD is, the higher the trade-offs between the paired indicators is. Following the research of Zeng et al. (2019), the average magnitude of trade-offs between paired indicators is defined as high for an RMSD which is larger than 0.2, low or synergistic when RMSD is smaller than 0.1, and moderate when RMSD is between 0.1 and 0.2.

3. Empirical results

3.1. Data description

This article uses data from 180 countries for the study. The countries list is based on Global Forest Product Model (GFPM2017; Buongiorno et al., 2003) and countries are classified by region according to this model. Data resources are the FAOSTAT^a and GFPM2017 is used to calculate the data needed.

3.1.1. Resources data

The carbon stock, forest stock and forest area are not available for every year. The data is published for every five years. Therefore, according to the calculation of Geng et al., 2017, the data of carbon stock, forest stock and the area is deduced from 1990 to 2015. The data is incomplete before 1990 and the latest data is until 2015. The FAO Global Forest Resources Alsaleh and Abdul-Rahim, 2018 (Feng et al., 2017) was used to prepare the input data.

3.1.2. Production data

An empirical analysis is carried out using the productions of the forestry, the raw materials which include fuelwood, industrial roundwood, and other industrial roundwood. This paper supposes bioenergy is fuelwood, the forestry productions are industrial roundwood and other industrial roundwood. According to the foregoing model construction, the quantity of productions and prices are all needed. The GFPM is an economic model of global production, consumption and trade of forest products. The original formulation and several applications are described in Buongiorno et al. (2003). From Geng et al., 2017 output file, the data that defines the demand equation for each country and end product in the base year can be obtained. In Geng et al., 2017, it contains the price of different products, the quantity demanded at that price. This was calculated based on GFPM2017 from 1992 to 2015.

According to the collected and calculated data of fuelwood production, biomass and carbon stock in living biomass of forest from 1990 to 2015, the average values per hectare globally in 180 regions are shown in above Fig. 2. Since the data of different product prices and the quantity demanded at that price was calculated based on GFPM2017 from 1992 to 2015, this paper used data from 1992 to 2015.

3.2. Carbon stock versus carbon flow

Considering the carbon sinks and carbon flux, Fig. 3 presents the annual harvested carbon about fossil fuel substitution of bioenergy and carbon sinks of different management strategies in 180 countries. The bioenergy production is about 274 m³/ha/year and average carbon stock is 98.6 tC/ha in Egypt. The forest area in Egypt is low and the density calculated from the wood fuel data estimated by FAO is relatively high. In order to present the data of other countries clearly, the figure cannot show the point of Egypt. The amount of harvested forest volume for bioenergy ranges from 0 to 50 m³/ha/year except Egypt and the carbon stock from 0 to 300 tC/ha/year.

According to earlier studies about the bioenergy conversion factors and equations (1)–(3), the results can be discussed with different combinations of the two factors. Mitchell et al. (2012) evaluated the average bioenergy conversion factor for woody biomass estimated is to be 0.51, meaning that harvesting 1 Mg of biomass C for bioenergy production will substitute for 0.51 Mg

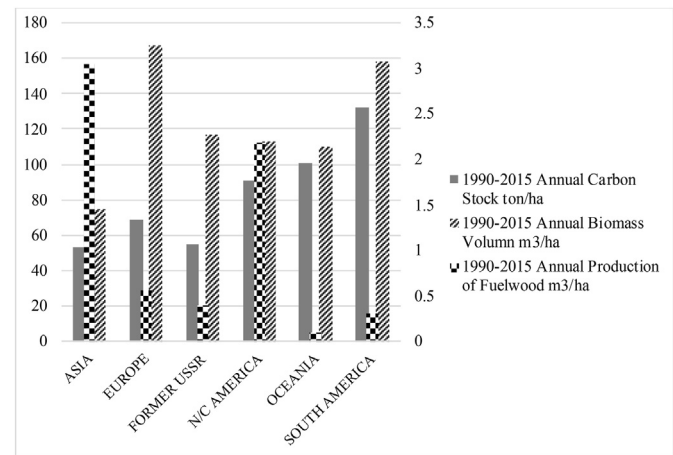


Fig. 2. Average Value of fuelwood, standing biomass and carbon stock in different regions.

fossil fuel C since less energy per unit C emissions is obtainable from biomass compared to fossil fuel. And it is assumed that 3.78 m³ wood substitute for 1 ton coal with saving 0.707 ton of carbon emissions (van Kooten et al., 1999). Based on the assumption and according to the collected carbon stock data and fuelwood production data of 180 countries in the world, the results of calculating the trade-offs of carbon sinks and flow is the average value from 1992 to 2015 which is shown in Fig. 4 (See the RMSD1 calculation results of 180 countries in Appendix).

According to the calculation results, the magnitudes of trade-offs about carbon sinks and carbon flow in 180 countries are mainly high for the value of RMSD being between 0.4 and 0.5. And the trade-offs are moderate in Iran (0.1299) and French Polynesia (0.1808) when RMSD is between 0.1 and 0.2. Furthermore, the calculation results are divided according to the intercontinental regions. The results of the calculations are summarized as shown in Table 1. From the calculation of the average value, it can be found that the results are similar in different intercontinental regions. It can be seen from the calculation results of the standard deviation that the deviation of trade-offs results in South America is the smallest, and the magnitude of trade-offs in Oceania is the most dispersed. From the comparison of the maximum and minimum values, it can be found that the extreme difference is largest in Asia, the minimum is 0.1299 in Iran, and the maximum is 0.5024 in the Republic of Korea. The smallest extreme difference is in Former USSR.

3.3. Economic benefit versus ecological benefit

According to the above analysis of forest economic benefit and ecological benefit, the relevant data collected from 180 countries in the world are brought into the model to calculate the total income. Fig. 5 presents the annual harvested carbon about fossil fuel substitution of bioenergy and forest biomass density in 180 countries.

Vass and Elofsson (2016) concluded the carbon content of wood is 1.459 tonnes CO₂/m³ with temperate forest and 0.912 tonnes CO₂/m³ with the boreal forest. The fossil fuel substitution coefficients are 0.138 tonnes C/m³ and 0.221 tonnes C/m³ of temperate forest and boreal forest which means bioenergy replacing coal in CHP plants, implying an emission offset. Obersteiner et al. (2006) proposed bioenergy production cost vary between 5 and 95 US\$/tC. Since the data of the bioenergy cost and other forestry products cost is unavailable in globally 180 countries. The cost is ignored in the empirical analysis. The global bioenergy

^a Website of Food and Agriculture Organization of the United Nations. <http://www.fao.org/faostat/en/#data>.

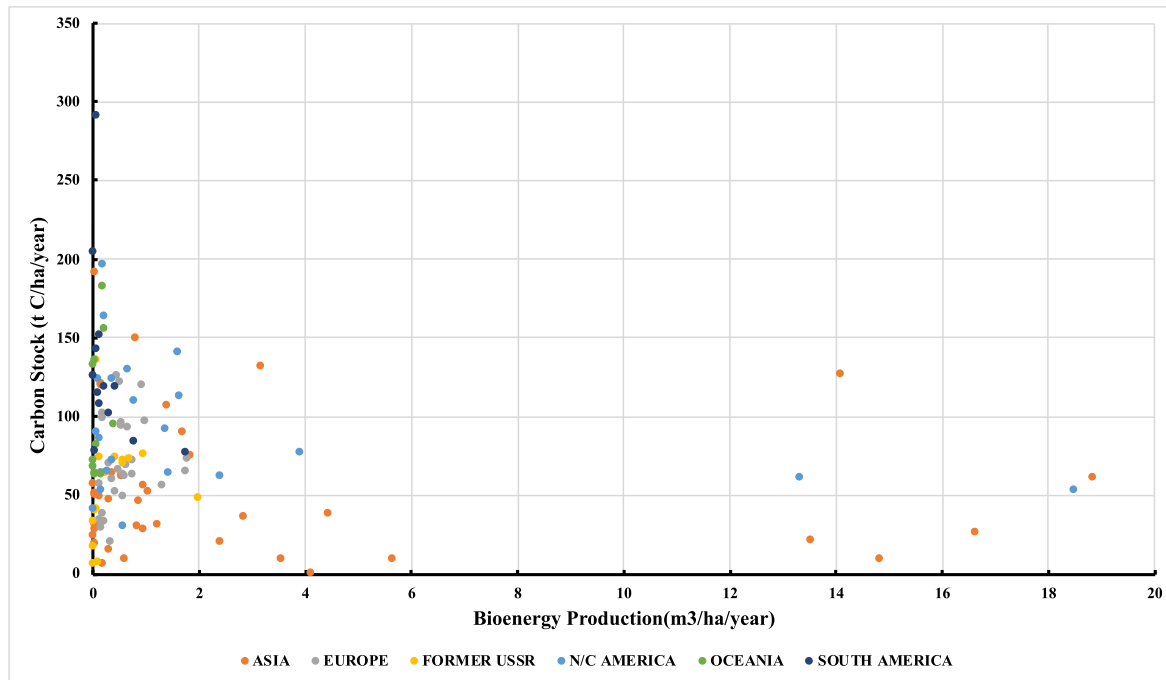


Fig. 3. The carbon stock and the bioenergy production of 180 countries.

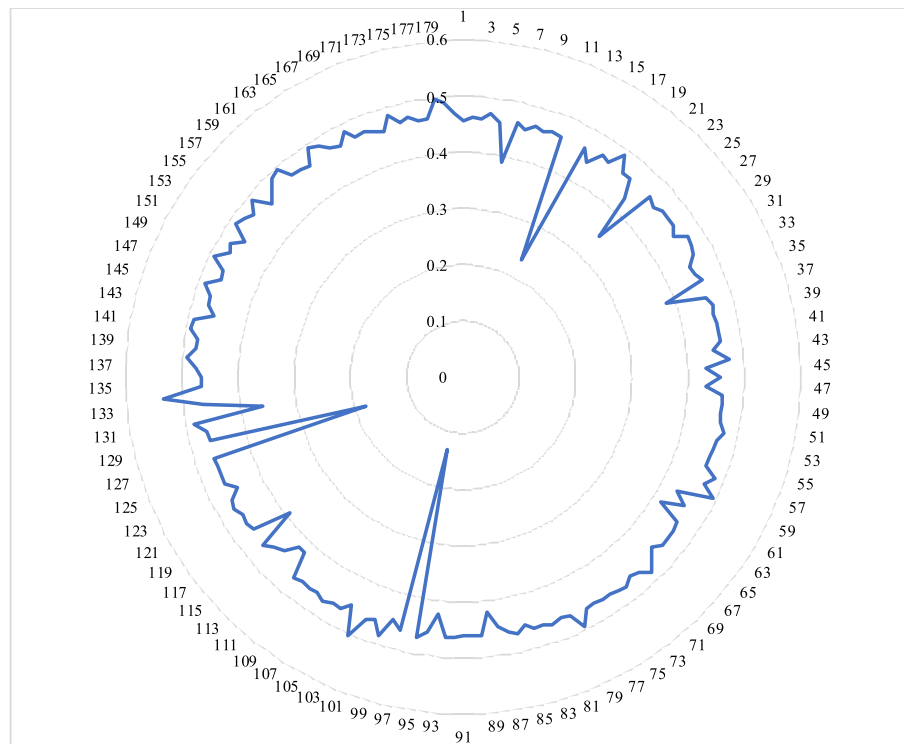


Fig. 4. The results of the magnitude of trade-offs between carbon sinks and flow.

price is between 23 and 62 US\$/m³ during 1992–2015, the average value is 42.875 US\$/m³. The data of carbon price is collected from [Sohngen et al. \(2017\)](#). It is converted to the 2010 price based on the 1992–2015 deflator, thus the average value from 1992 to 2015 is 27.52 US\$/tC. According to the average value of earlier research and based on equation (9), the economic benefit and ecological benefit

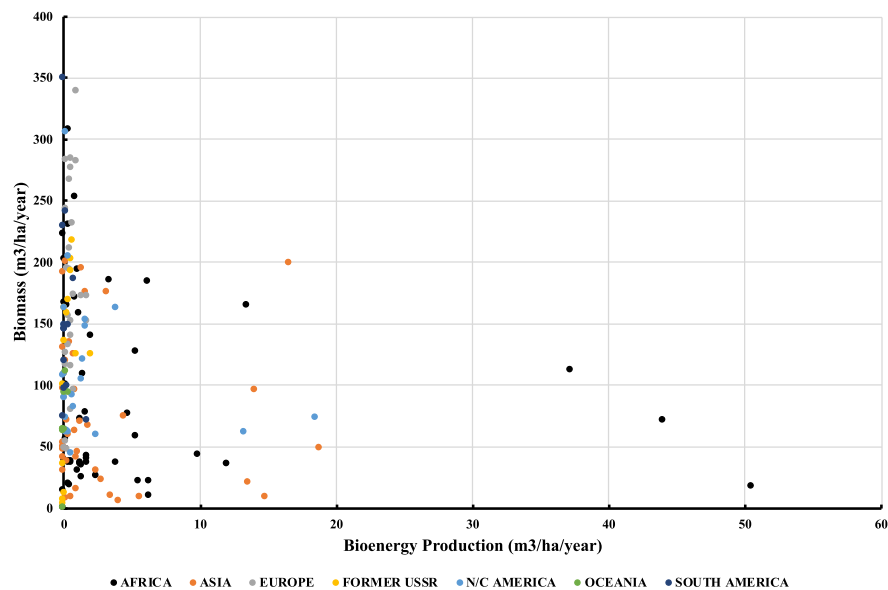
of forest per unit area in various countries around the world can be calculated. Then the magnitude of the trade-offs between the forest economic benefit and ecological benefit is shown in Fig. 6 (See the RMSD2 calculation results of 180 countries in Appendix).

The results of the calculations are summarized as shown in Table 2. From the perspective of the average, it can be found that

Table 1

Characteristics of the magnitude of trade-offs between carbon sinks and flow.

Region	Average Value	Standard Deviation	Maximum Value	Minimum Value
Africa	0.4467	0.0426	0.4892	0.2337
Asia	0.4480	0.0556	0.5024	0.1299
Europe	0.4771	0.0122	0.4956	0.4551
Former USSR	0.4737	0.0110	0.4974	0.4602
North America	0.4628	0.0162	0.4929	0.4159
Oceania	0.4374	0.0939	0.5329	0.1808
South America	0.4669	0.0094	0.4951	0.4534

**Fig. 5.** The standing biomass and the bioenergy production of 180 countries.

the magnitude of trade-offs in South America is the largest, results of the other regions are almost similar. According to the standard deviation, the countries in South America are highly dispersed, and the magnitude of trade-offs is concentrated in Former USSR. In the same way, the differences between the extreme values are large in South America. The largest is 0.6808 in South America which is in Ecuador, the minimum is 0.2245 in Uruguay. The lowest trade-offs between forest economic benefit and ecological benefit is 0.0649 which is in Iran.

4. Analysis

4.1. Global analysis

According to the *Global Forest Resources Assessment* [Rabbani et al., 2018](#), the global forest area is 3.99 billion hectares, with a coverage rate of 30.6%, a per capita area of 0.6 ha, a forest volume of 431 billion cubic meters, and a biomass carbon reserve of about 250 billion tons (including above-ground and underground biomass). The distribution of forest resources varies from different regions with 1.015 billion hectares in Europe, 842 million hectares in South America, 751 million hectares in Central and North America, 624 million hectares in Africa, 593 million hectares in Asia and 174 million hectares in Oceania. The top five countries with the richest forest resources are Russia, Brazil, Canada, the United States, and China, which together account for 54% of the total global forest area. In the use of bioenergy in countries around the world, fuel-wood accounts for a large proportion of the total energy

consumption of developing countries, especially in African countries, while the proportion of total energy consumption in developed countries is still small. However, all the countries are increasingly aware of the environmental benefits of bioenergy.

From a global perspective, the average RMSD in 180 countries is shown in the following [Fig. 7](#) from 1992 to 2015. Where RMSD1 represents the trade-offs between bioenergy production and carbon sinks measured from the perspective of carbon sinks and carbon flow, and RMSD2 is a trade-off between forest economic benefits and ecological benefits. The higher value implies there is a trade-off between the two indicators and the lower value represents the synergistic of the two. It can be found that the two measures show a consistent trend, with a large change in 2011, 2012 and 2013. From the perspective of carbon stock change, the carbon stocks remained relatively stable from 1992 to 2015. In 2011, the global carbon storage density showed a significant decline. In 2012, the carbon storage density rebounded, and the carbon storage density declined again in 2013. As forest management and timber deforestation cause large changes in carbon stock density, countries also need to balance the production of forest products and the formation of carbon sinks to achieve coordinated development between the two.

4.2. Regional analysis

According to [Table 3](#), it concludes the characteristics of countries with the highest value of trade-offs and lowest trade-offs from two perspectives on carbon reductions and forest incomes in different

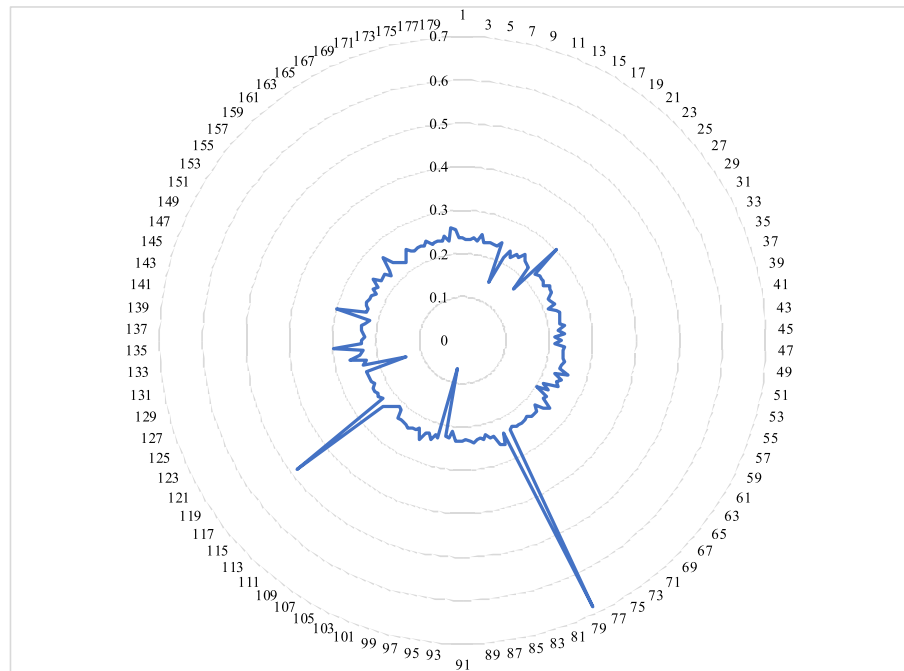


Fig. 6. The results of trade-offs between forest economic benefit and ecological benefit.

Table 2

Characteristics of the magnitude of trade-offs between economic benefit and ecological benefit in different regions.

Region	Average Value	Standard Deviation	Maximum Value	Minimum Value
Africa	0.2283	0.0197	0.3033	0.1479
Asia	0.2326	0.0498	0.4859	0.0649
Europe	0.2363	0.0147	0.2997	0.2193
Former USSR	0.2391	0.0086	0.2603	0.2306
North America	0.2338	0.0119	0.2550	0.2033
Oceania	0.2336	0.0387	0.3004	0.1361
South America	0.2705	0.1237	0.6808	0.2245

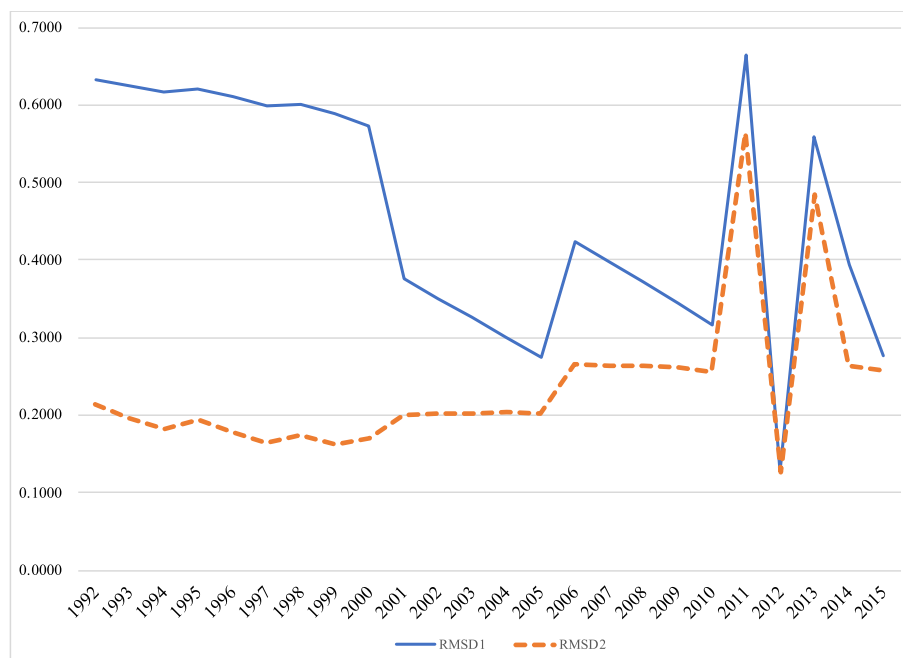


Fig. 7. The magnitude of trade-offs between bioenergy production and carbon sinks from 1992 to 2015.

regions.

Among the calculations of trade-offs from carbon reductions and forest benefit in all countries in the African region, the region with the lowest RMSD is Djibouti in terms of the trade-offs between carbon sinks and carbon flow, and the region with the lowest RMSD is Egypt which is the trade-offs between forest economic benefit and ecological benefit. Djibouti is poor in forest resources and has fewer forest products. It has a small amount of fuelwood, and its forest carbon storage per unit area kept steady. Hence there are more synergistic between bioenergy production and carbon sinks. Carbon storage in the Egypt forest is the 15th in Africa. In forestry management, its amount of fuelwood per unit area is relatively high and most forest products are used for direct combustion to generate energy. Although it had the lowest magnitude of trade-offs between economic benefit and ecological benefit, it also had large trade-offs since the value is more than 0.2.

From the two perspectives of trade-offs between bioenergy production and carbon sinks in the Asian region, Iran is the country with the lowest value in Asia, even in the world. Iran is rich in natural resources. The main industry of the local economy is oil exploitation, and other industries are underdeveloped. There are rich agricultural resources and forestry resources. However, Iran has 10% of the world's oil resources and the amount of deforestation is relatively small. Its development didn't rely on bioenergy production. Forestry is part of the jurisdiction of the Ministry of Agriculture. Therefore, there is more coordinated development between bioenergy production and carbon sinks.

In Europe, North America, South America, and former Soviet Union countries, the maximum and minimum values of the trade-off measurement results are large, indicating that there is a large trade-off between bioenergy production and carbon sinks in the entire region. Denmark measures the largest of its trade-offs in terms of carbon sinks and carbon flows, meanwhile, Denmark is the smallest in Europe when considering the trade-offs from the perspective of forest economic benefits and ecological benefits. Denmark's natural resources are scarce. Except for oil and natural gas, there are few other mineral deposits. All coal, iron and other minerals are imported. Denmark's oil reserves on the North Sea shelf are estimated at 290 million tons and natural gas reserves are about 200 billion cubic meters. The forest covers an area of 486,000 ha, with a coverage rate of about 10%. Therefore, the magnitude of trade-offs between carbon sinks and carbon flow is larger and trade-offs between economic benefit and ecological benefit is lower, although the two values are all high.

In Oceania, French Polynesia is the country with the lowest value from the two perspectives of trade-offs between bioenergy production and carbon sinks. The main economic industry in French Polynesia is tourism, and thermal power generation is the main energy source. The carbon stocks and fuelwood in the forest area are not the highest, but there are more synergistic between the harvest for bioenergy and carbon sinks. When the forestry revenues from carbon sequestration and forest products are both considered, it also presented the most synergistic in Oceania.

From the global intercontinental comparison, it can be found that countries with lower magnitude trade-offs usually had a small amount of harvest. The use of fuelwood as bioenergy is also less. The role of forests in the environment is mainly due to the carbon sequestration of growing trees. The production and use of bioenergy can replace fossil fuels. However, due to the current low efficiency of substitution, the amount of carbon dioxide reduction is small. It showed that reducing the total amount of carbon dioxide in the air, the strategies should be, reducing deforestation, strengthening forest management, and increasing the amount of forest volume per unit area. The measurement of trade-offs between economic benefit and ecological benefit showed that countries with lower magnitude of trade-offs usually had higher carbon reserves and higher forest harvesting, such as Denmark. Therefore, forest management should simultaneously enhance its commercial value development and ecological environment value.

5. Conclusions and policy implications

Forests are important economic assets and huge environmental assets. This study focuses on the analysis of forests fixing carbon dioxide, bioenergy substituted fossil fuels for carbon dioxide emissions reductions and forest product production. Hence, the relevant conclusions can be summarized into the following policy implications.

First, developing forestry can bring economic income to areas that are rich in forest resources and at the same time it can be able to mitigate climate change when emitting carbon dioxide were charged. It is an effective way for countries to pursue green economic development based on balancing the bioenergy production and carbon sinks.

Second, partition management is available in forest management and it could fully realize the economic and ecological functions of forests. It can delineate a part of forest areas for forest products, bioenergy, and other production activities to maximize economic returns. Meanwhile, forest carbon sinks area should maximize the carbon stocks of the forests according to the needs of tree growth. The protected areas are the protection of some precious forestry species in the area and prohibit any commercial development activities. It is used in countries such as the United States and Canada, mainly through the division of forestry areas into timber production areas, carbon sink forest areas, ecological forestry protection areas and other areas.

Third, forest carbon sinks should be considered in environmental governance. The current environmental damage has not been valued which leads many countries, especially developing countries, to develop agriculture at the expense of forest resources. Countries should give full attention to forestry in combating climate change and integrate forestry into climate change mitigation.

Finally, when promoting energy policies to reduce carbon emission such as developing renewable energy, considering whether the carbon sequestration function of local forests could achieve the carbon emission reduction targets of the region. It could achieve coordinated development between economy and ecology. For example, oil supports the national energy supply in

Table 3
Countries with the larger or lower magnitude trade-offs in different regions.

Region	Countries with larger value	Countries with lower value
Carbon stock versus carbon flow (RMSD1)		
Africa	Liberia (0.3033)	Djibouti (0.1479)
Asia	Singapore (0.4859)	Iran (0.0649)
Europe	Denmark (0.2997)	Czech Republic (0.2193)
Former USSR	Turkmenistan (0.2603)	Kyrgyzstan (0.2306)
North America	Saint Lucia (0.2550)	Haiti (0.2033)
South America	Ecuador (0.6808)	Uruguay (0.2245)
Oceania	Tonga (0.3004)	French Polynesia (0.1361)
Economic benefit versus ecological benefit (RMSD2)		
Africa	Ghana (0.4892)	Egypt (0.2337)
Asia	Korea (0.5024)	Iran (0.1299)
Europe	Slovenia (0.4956)	Denmark (0.4551)
Former USSR	Turkmenistan (0.4974)	Kyrgyzstan (0.4602)
North America	Dominican Republic (0.4930)	Haiti (0.4159)
South America	Ecuador (0.4951)	Uruguay (0.4534)
Oceania	Tonga (0.5329)	French Polynesia (0.1808)

Note: The numerator is the result of the trade-offs measurement in the region in the brackets behind countries' name.

Iran and the forest provides carbon sequestration without a harvest for bioenergy. There is a synergistic relationship between bioenergy production and carbon sinks in Iran.

This article calculated and analyzed the trade-offs between bioenergy production and carbon sinks in 180 countries around the world. The shortcomings are that the global unified parameters adopted by some parameters cannot accurately capture the characteristics of each region. Due to the current problem, further research on forest characteristics in different regions needs to be done.

Declaration of competing interest

The authors declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service or company that could be construed as influencing the position presented in the manuscript entitled “*To harvest or not to harvest? Forest management as a trade-off between bioenergy production and carbon sink*”.

CRediT authorship contribution statement

Boqiang Lin: Conceptualization, Methodology, Software, Data curation, Writing - original draft. **Jiamin Ge:** Methodology, Software, Data curation, Writing - original draft.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2020.122219>.

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